Synchrotron Emission

The relativistic spectrum of synchrotron radiation peaks at

$$\nu_c = \gamma^2 \nu_g ,$$

where ν_g is the non-relativistic synchrotron frequency

$$\nu_g = \frac{qB}{2\pi m}$$

$\therefore$ If we can estimate B, we can estimate the Lorentz factor γ:

$$\gamma = \sqrt{\frac{\nu_c}{2.8\times10^6 B}}$$

by inserting constants.

Brightness Temperature

What temperature would a black body need to be in order to produce this brightness at this frequency?

$$\sigma_B = \frac{S_\nu c^2}{2k_B \Omega \nu^2} = \frac{S_\nu c^2 D^2}{2\pi k_B \nu^2 R^2}$$

as $\Omega = \frac{\pi R^2}{D^2}$

Minimum Energy

Each electron has $E_{individual} \propto \gamma \propto B^{-1/2}$ for a fixed ν.
From Larmor's formula: $P \propto B^2\gamma^2 \propto B$
For N electrons, total power radiated is $L \sim N P_{individual}$
So if observed luminosity is unchanged then $N \propto P^{-1} \propto B^{-1}$

$$\text{So} \quad E_e = N E_{\text{individual}} \propto B^{-1} B^{-1/2} \propto B^{-3/2}$$

Energy stored in magnetic field $E_B \propto B^2$

Hence $\frac{d}{dB}(E_e + E_B) = -\frac{3}{2}B^{-5/2} + 2B = 0$

So $-\frac{3}{2}E_e + 2E_B = 0$

$\therefore \quad E_B = \frac{3}{4}E_e$

This is the magnetic field that corresponds to the minimum energy.

power law index, p, of the electron energy distribution describes how quickly the number of electrons decreases as their energy ϵ increases:

$$n(\epsilon) \propto \epsilon^{-p}$$

spectral index, α, is related to the power law index by

$$\alpha = -\frac{p-1}{2}$$

putting together we know $\mathcal{E}_{electron} = \gamma m_e c^2$

and that $\nu \propto B\gamma^2$

so $\mathcal{E}_{electron} \propto \nu^{1/2}$

The power radiated by electrons is $P_{electron} \propto \mathcal{E}^2$ (at fixed B)

The flux per unit frequency, F_ν

$$F_\nu d\nu \propto n(\varepsilon) P(\varepsilon) d\varepsilon$$

Then $\frac{d\mathcal{E}}{d\nu} \propto \mathcal{E}^{-1}$

So:

$$F_\nu \propto \nu^{(1-p)/2} \propto \nu^\alpha$$

Derivation of $E_{\rm min} \approx 8 \times 10^6 \eta^{4/7} V^{3/7} \nu^{2/7} L_\nu^{4/7}$

$$E_{\rm total} \propto \underbrace{\eta L_\nu \nu^{1/2} B^{-3/2}}_{\text{particles}} + \underbrace{V B^2}_{\text{magnetic field}}$$

particles:

$$\epsilon_{\rm electron} \propto \nu^{1/2} B^{-1/2}, \qquad P_{\rm electron} \propto B^2 \epsilon^2 \propto B\nu.$$

The luminosity in a fixed fractional frequency band is proportional to νL_ν

Therefore

$$N \propto \frac{\nu L_\nu}{P_{\rm electron}} \propto \frac{\nu L_\nu}{B\nu} = \frac{L_\nu}{B}$$

$$E_{\rm particles} \propto \eta N \epsilon \propto \boxed{\eta L_\nu \nu^{1/2} B^{-3/2}}$$

minimum:

$$\frac{{\rm d}E_{total}}{{\rm d}B} = 0 \quad \Rightarrow \quad \eta L_\nu^{1/2} B^{-5/2} \propto VB$$

Rearranging gives $B_{eq} \propto \eta^{2/7} L_\nu^{2/7} \nu^{1/7} V^{-2/7}$

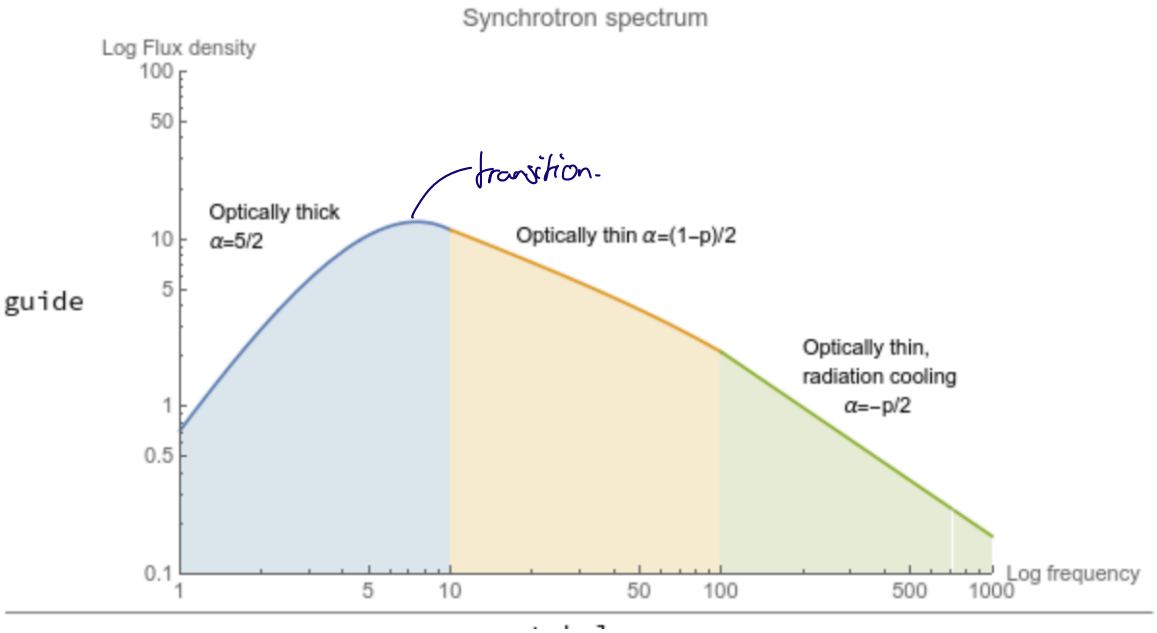
$$\boxed{E_{\rm min} \propto \eta^{4/7} V^{3/7} \nu^{2/7} L_\nu^{4/7}}.$$

Synchrotron Emission from power-law distribution of electrons

- Self Absorbed part has spectral index, $\alpha = +2.5$ (Thermal emission has $\alpha = +2$)
- Optically thin branch has spectral index directly related to the underlying power law spectrum of electron distribution
- Radiation cooling is not often seen which means you can directly calculate $p = 1 - 2\alpha$ as given $S_\nu \propto \nu^\alpha$:

$$\alpha = \frac{\log\left(S_{\nu_1}/S_{\nu_2}\right)}{\log\left(\nu_1/\nu_2\right)}$$

Note: Sometimes the convention of $S_\nu \propto \nu^{-\alpha}$ is used...



For the next part we ignore the constants and just work in proportionality...

We define

$$\tau_\nu = \kappa_\nu s$$

where: κ_ν is the absorption coefficient
s is the depth along line of sight.

$\tau_\nu \ll 1$: optically thin
$\tau_\nu \gg 1$: optically thick
$\tau_\nu \sim 1$: transition.

Absorption coefficient: $\kappa_\nu \propto N_0 B^{(p+2)/2} \nu^{-(p+4)/2}$

Emission coefficient: $\epsilon_\nu \propto N_0 B^{(p+1)/2} \nu^{(1-p)/2}$

The source function S_ν (not to be confused with S_ν for flux density) is given by:

$$\mathcal{S}_\nu = \frac{\epsilon_\nu}{\kappa_\nu} \propto B^{-1/2} \nu^{5/2}$$

The specific intensity observed from a region of physical depth s and no background illumination is

$$\boxed{I_\nu = \mathcal{S}_\nu \left(1 - e^{-\tau_\nu}\right)}$$

For very small $\tau_\nu \ll 1$: $1 - e^{-\tau_\nu} \approx \tau_\nu$

and $\tau_\nu = \kappa_\nu s \quad \therefore \quad I_\nu = \epsilon_\nu s \propto \nu^{(1-p)/2}$ — so slope gives p.

For very large $\tau_\nu \gg 1$: $I_\nu = S_\nu \propto \nu^{5/2}$ — tells you nothing directly.

We define $\tau_{\nu_1} = 1$

Then introduce dimensionless frequency $z = \dfrac{\nu}{\nu_1}$

$$\mathcal{S}(\nu) \propto \nu^{5/2} \qquad \therefore \mathcal{S}(\nu) = \mathcal{S}(\nu_1) z^{5/2}$$

$$\tau_\nu \propto \kappa_\nu \propto \nu^{-(p+4)/2} \qquad \therefore \tau_\nu = z^{-(p+4)/2}$$

It hence follows $\boxed{I_\nu = \mathcal{S}_{\nu_1} J(z, p)}$

where $J(z,p) = z^{5/2}\left(1 - \exp\left[-z^{-(p+4)/2}\right]\right)$

To obtain the observed flux density we integrate over the source's apparent solid angle... For this uniform source model:

$$F_\nu = \Omega I_\nu = \frac{\pi R^2}{D^2} I_\nu$$

↑ solid angle.

As $I_\nu = \mathcal{S}_{\nu_1} J(z,p)$:

Optically thick: $\tau_\nu \gg 1 \Rightarrow J(z,p) \approx z^{5/2}$

$$I_\nu = S_{\nu_1} z^{5/2} = S_\nu$$

So $$I_\nu \propto B^{-1/2} z^{5/2} \equiv B^{-1/2} \left(\frac{\nu}{\nu_1}\right)^{5/2}$$

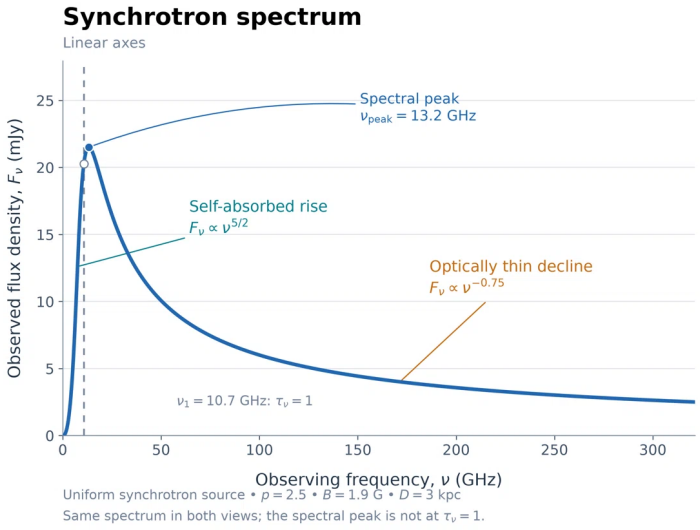
Optically thin: $\tau_\nu \ll 1 \Rightarrow J(z,p) \approx z^{5/2} \tau_\nu \equiv \kappa_\nu s$

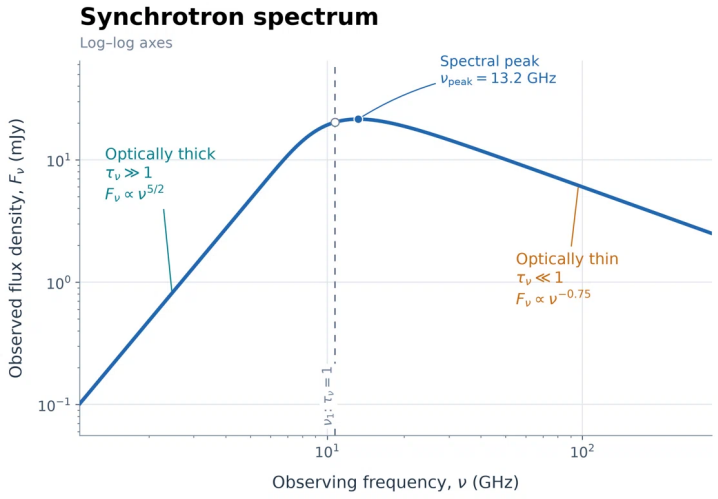
$$\therefore I_\nu \approx \mathcal{S}_\nu \kappa_\nu s \equiv \frac{\epsilon_\nu}{\kappa_\nu} \kappa_\nu s = \epsilon_\nu s$$

$$\therefore I_\nu \propto N_0 B^{(p+1)/2} \nu^{-(p-1)/2} s$$

$$F_{\nu,\text{thick}} \propto \frac{\pi R^2}{D^2} B^{-1/2} \nu^{5/2}$$

$$F_{\nu,\text{thin}} = \frac{\pi R^2 s}{D^2} \epsilon_\nu \equiv \frac{V}{D^2} \epsilon_\nu \propto N_0 B^{(p+1)/2} \nu^{-(p-1)/2}$$



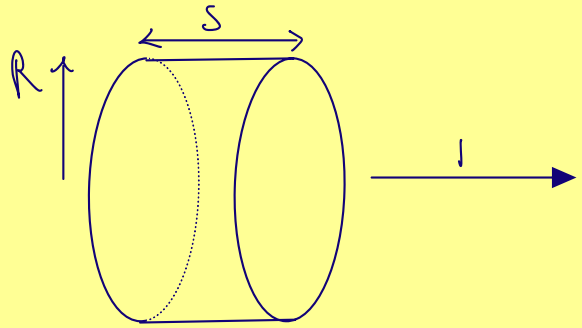


f is the volume filling factor: fraction of assumed spherical source that contains emitting plasma.

$$\therefore \quad V_{\text{emitting}} = \frac{4}{3}\pi R^3$$

We introduce the following construction from Pacholczyk:

- Radiation-transfer eq. is simplest when every line of sight passes through the same depth.
- A sphere does not have constant depth.
- We convert to a uniform slab with a circular face ie. a cylinder viewed along its axis.



$$\frac{4}{3}\pi R^3 f = \pi R^2 s$$

$$\therefore \quad \boxed{s = \frac{4}{3} f R}$$

The cylinder preserves:

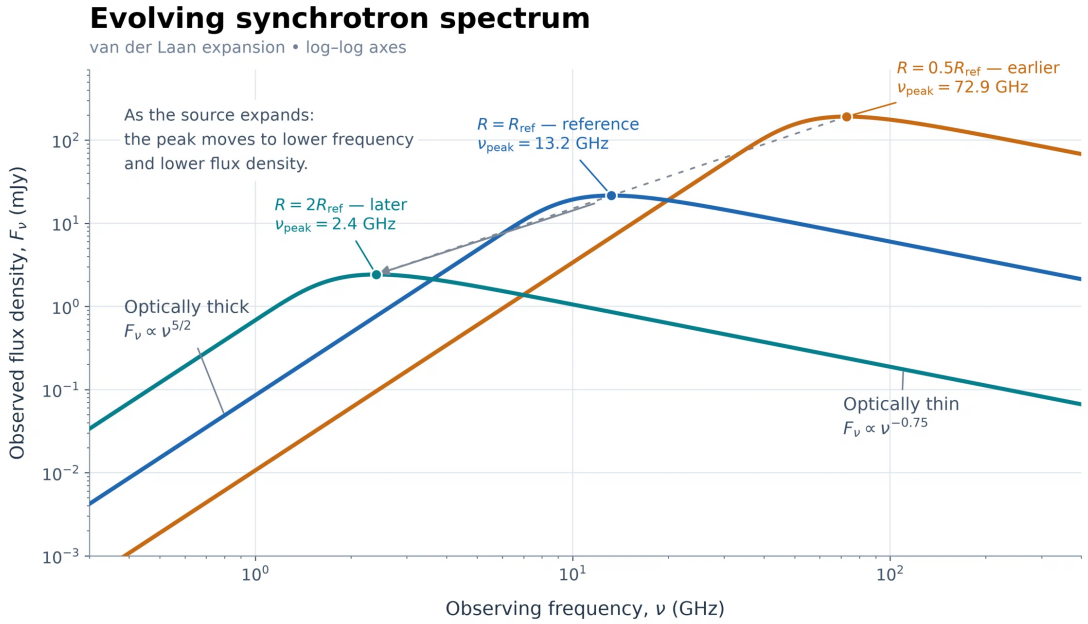
1. apparent area on the sky πR^2
2. Total emitting volume $\frac{4}{3} f \pi R^3$

Time Evolution of Synchrotron Emission

This model considers a synchrotron emitting blob which is expanding.

Due to adiabatic losses, the number density $N_0 \propto R^{-(2+p)}$

The magnetic field is assumed to evolve according to magnetic flux conservation - ie. $B \propto R^{-2}$



Expansion model: $B \propto R^{-2}$, $N_0 \propto R^{-(p+2)}$; $p = 2.5$ stays fixed.

At fixed frequency: thick emission rises as R^3; thin emission fades as R^{-5}.

Reference: $B_{ref} = 1.9$ G, $D = 3$ kpc. Markers show spectral peaks, not $\tau_\nu = 1$.

Observing Simulation

See notebook but essentially it does the following:

Assume source expands at constant speed v (code assumes $v = c/\sqrt{3}$) such that

$$R(t) = R_0 + vt$$

The apparent area increases as

$$\Omega(t) = \frac{\pi R(t)^2}{D^2}$$

Van der Laan assumptions:

$$B(t) = B_0 \left(\frac{R(t)}{R_0}\right)^{-2} \quad \& \quad N_0(t) = N_{00} \left(\frac{R(t)}{R_0}\right)^{-(p+2)}$$

- The field weakens as the source expands.
- The electron distribution changes because the particles become more dilute and loose energy through adiabatic expansion.

Recall: N_0 is the normalisation of the energy distribution